Lightweight Federated Learning with Differential Privacy and Straggler Resilience

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Privacy Concerns in Federated Learning (FL)

- Federated Learning (FL) enables collaborative model training while keeping raw data local.
  - Clients compute gradients/parameters locally and send them to the server for aggregation.

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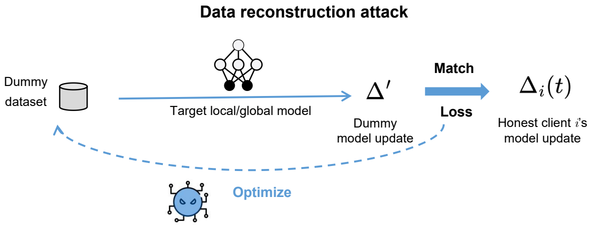
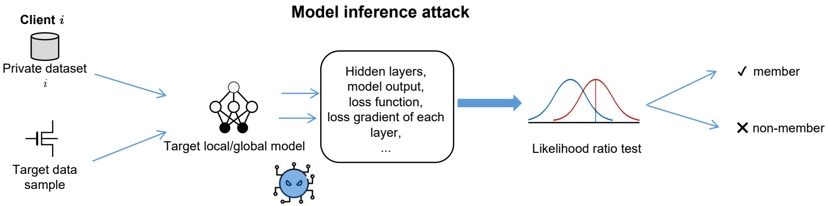
  - Clients compute gradients/parameters locally and send them to the server for aggregation.
- **FL alone is not private enough.**
  Raw data or membership can be inferred from exchanged parameters.

Examples of Inference Attacks





Existing Privacy-Preserving FL

- **Differential Privacy (DP)** offers rigorous guarantees against various unknown attacks by making inputs indistinguishable from observed target model parameters.
  - **Conventional Local Differential Privacy (DP)**:
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    - *Problem*: Accumulated noise during server-side aggregation leads to low training **accuracy**.

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  - SMPC: Individual client-to-server updates are secured (e.g., via pairwise encryption, homomorphic encryption, coded computation).
  - SMPC+DP: Small additional noise for DP on the sum.
  - *Problems*:
    - High communication and computation **overheads**.
    - Vulnerability to **stragglers** (client-to-server or client-to-client links).

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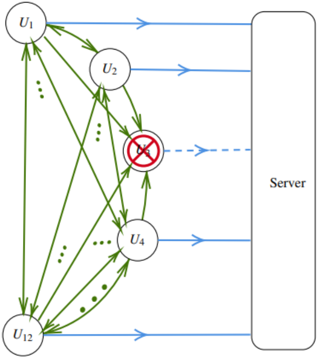

Specific SMPC Limitations

- **Straggler Issues**:
  - Pairwise encryption is highly sensitive to stragglers.
  - Some ”straggler-resilient” methods rely on impractical assumptions about consistent peer communication, making them vulnerable in practice.
- **Computation/Communication Overheads**:
  - Homomorphic Encryption (HE): Large ciphertext size, extra decryption rounds.
  - SecAgg: Large encrypted parameters, extra overhead for straggler handling via secret sharing in every round.

Objective

Privacy Mechanism	DP-private	Straggler-resilient	Comm./Comp. Overhead	Accuracy
Vanilla local DP noise	Yes	Yes	Low	Low
HE + DP noise	Yes	Yes	High	High
Coded Computation (e.g., SwiftAgg+)	No[1]	No	Medium	High
Pairwise Encryption	No[1]	No	Low	High
SecAgg	No	No	High	High
Dordis (SecAgg+DP noise)	Yes	No	High	High
Our Scheme (LightDP-FL)	**Yes**	**Yes**	**Low**	**High**

***Open Challenge**: Develop an efficient, lightweight FL mechanism with strict DP, straggler resilience, and low overheads.*

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- The SMPC schemes with 'fully secure' client-to-server exchanges come with significant overheads and straggler vulnerability.
- Whether do we really need the individual client-to-server updates to be fully safe with those complicated SMPC schemes?
- DP is enough.
- **Objectives**:
  - Provable Differential Privacy against untrusted peers and server.
  - Straggler resilience.
  - Low communication and computation overheads.
  - High training accuracy.

System Model

- Training objective:

$$\omega^* = \arg\min_{\omega} \left\{ F(\omega, \mathcal{D}) \triangleq \sum_{i \in \mathcal{N}} \frac{|\mathcal{D}_i|}{\sum_{i' \in \mathcal{N}} |\mathcal{D}_{i'}|} F_i(\omega, \mathcal{D}_i) \right\}.$$

- Local model update:

$$\omega_i^{(t)} := \omega_i^{(t)} - \eta \nabla F_i(\omega_i^{(t)}; \xi_i^{(t)}).$$

- Global aggregation:

$$\omega^{(t)} = \frac{\sum_{i \in \mathcal{N} \setminus \mathcal{N}_S^{(t)}} \tilde{\omega}_i^{(t)}}{N - S^{(t)}},$$

with **Straggler Set:** $\mathcal{N}_S^{(t)} \subset \mathcal{N}$ at round t and $|\mathcal{N}_S^{(t)}| = S^{(t)} \leq \bar{S}$.

Attack Model

Colluder Set: $\mathcal{N}_{\mathrm{C}}^{(t)} \subset \mathcal{N}$ at round t with $|\mathcal{N}_{\mathrm{C}}^{(t)}| = C^{(t)} \leq \bar{C}$

- Attack model: semi-honest/honest-but-curious
  - All parties honestly follow the training protocol.
  - An adversary may collude with the untrusted server and a subset of colluding clients.
  - Unknown attacks.

Privacy Metric: DP

Definition of Differential Privacy

A randomized algorithm $\mathcal{M}$ is (ε, δ)-differentially-private if for all adjacent datasets $\mathcal{D}, \mathcal{D}'$ differing by at most one data sample:

$$\Pr[\mathcal{M}(\mathcal{D}) \in \mathcal{S}] \leq e^{\varepsilon} \Pr[\mathcal{M}(\mathcal{D}') \in \mathcal{S}] + \delta,$$

where:

- ε controls the distinguishability between outputs of $\mathcal{M}(\mathcal{D})$ and $\mathcal{M}(\mathcal{D}')$.
- δ represents the probability that the privacy guarantee may fail.

DP Constraint

Following DP definition:

- Privacy loss function for a privacy mechanism $\mathcal{M}$:

$$\mathcal{L}_i(\mathcal{M}) = \ln \frac{\Pr(\mathcal{M}(\mathcal{D}))}{\Pr(\mathcal{M}(\mathcal{D}'))} \tag{1}$$

  - $\mathcal{M}(\mathcal{D})$ represents all outcomes of mechanism $\mathcal{M}$ based on dataset $\mathcal{D}$ during all training rounds.
  - $\mathcal{D}$ and $\mathcal{D}'$ are two adjacent datasets where only one data sample in client i's datasets $\mathcal{D}_i$ and $\mathcal{D}_i'$ differs.

- Rewrite (ε, δ)-DP guarantee:

$$\Pr(|\mathcal{L}_i(\mathcal{M})| \geq \varepsilon) \leq \delta. \tag{2}$$

Our Solution: LightDP-FL

- **Key Idea**:
  - Incorporate both *individual noise* (n_i) and *pairwise noise* (r_{ij}) into each client's parameter.
  - Utilize upper bounds on stragglers ($\bar{S}$) and colluders ($\bar{C}$) to prove sufficient noise variance for DP in the worst case.
  - Optimize the expected convergence bound by flexibly controlling noise variances.

LightDP-FL

Generation of Pairwise Noise

- Each pair of clients (i, j) agrees on a common seed s and a pseudorandom generator (PRG) (e.g., using Diffie-Hellman key agreement).
- This is a one-time communication before the learning process.
- The PRG generates pairwise noise $r_{ij}^{(t)} = r_{ji}^{(t)} \sim \mathcal{N}(0, \sigma_{ij}^2 \mathbf{I})$ for each round t.

LightDP-FL

Local Parameter Updates

- Client i masks local update ω_i with:
  - Individual Gaussian noise $n_i \sim \mathcal{N}(0, \sigma_i^2 \mathbf{I})$.
  - Sum of pairwise noises: $+\sum_{a>i} r_{ia}$ and $-\sum_{b<i} r_{bi}$.

$$\tilde{\omega}_i := \omega_i + \sum_{a \in \mathcal{N}: a>i} r_{ia} - \sum_{b \in \mathcal{N}: b<i} r_{bi} + n_i. \tag{3}$$

- Actual local disturbance for client i:

$$\begin{aligned} m_i &= \sum_{a: a>i, a \notin \mathcal{N}_C} r_{ia} - \sum_{b: b<i, b \notin \mathcal{N}_C} r_{bi} + n_i \\ &\sim \mathcal{N}\left(0, \sum_{j \neq i, j \in \mathcal{N} \setminus \mathcal{N}_C} \sigma_{ij}^2 \mathbf{I} + \sigma_i^2 \mathbf{I}\right). \end{aligned} \tag{4}$$

LightDP-FL

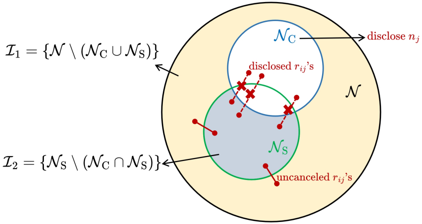
Global Aggregation

- After receiving all local updates $\tilde{\omega}_i^{(t)}$ from client $i \in \mathcal{N} \setminus \mathcal{N}_S^{(t)}$, the server updates the global parameter as:

$$\omega^{(t)} = \frac{1}{N - S^{(t)}} \sum_{i \in \mathcal{N} \setminus \mathcal{N}_S^{(t)}} \omega_i^{(t)} + n_D^{(t)}.$$

  where $n_D^{(t)}$ is the accumulated noise term, including the uncanceled pairwise and all individual noise terms.
- Further consider the colluder set to give the actual global disturbance.

Global Disturbance



- Due to the straggler issue, pairwise noise terms $r_{ij}, \forall i \in \mathcal{N} \setminus \mathcal{N}_\mathrm{S}, j \in \mathcal{N}_\mathrm{S}$ remain uncanceled in the aggregated noise term n_D.
- Remaining colluders $j \in \mathcal{N}_\mathrm{C} \setminus (\mathcal{N}_\mathrm{C} \cap \mathcal{N}_\mathrm{S})$ will disclose their individual terms n_j and pairwise terms r_{ij}'s.
- Only r_{ij}'s with $i \in \mathcal{I}_1 = \{\mathcal{N} \setminus (\mathcal{N}_\mathrm{C} \cup \mathcal{N}_\mathrm{S})\}$ and $j \in \mathcal{I}_2 = \{\mathcal{N}_\mathrm{S} \setminus (\mathcal{N}_\mathrm{C} \cap \mathcal{N}_\mathrm{S})\}$, and noise terms n_i's with $i \in \mathcal{I}_1$ remain in the global disturbance term for privacy.

DP Guarantee

- Privacy loss function for a privacy mechanism $\mathcal{M}$:

$$\mathcal{L}_i(\mathcal{M}) = \ln \frac{\Pr(\mathcal{M}(\mathcal{D}))}{\Pr(\mathcal{M}(\mathcal{D}'))} \tag{5}$$

  $\mathcal{M}(\mathcal{D}) = \{\tilde{\omega}_j^{(t)}, j \in \mathcal{N} \setminus \mathcal{N}_S^{(t)}; \omega^{(t)}, t \in \mathcal{T}\}$ represents all outcomes of mechanism $\mathcal{M}$ based on dataset $\mathcal{D}$ during all training rounds.

- Equivalent DP Guarantee:

$$\Pr(|\mathcal{L}_i(\mathcal{M})| \geq \varepsilon) \leq \delta.$$

Privacy Analysis

DP Guarantee for client $i \in \mathcal{I}_1 = \mathcal{N} \setminus (\mathcal{N}_C \cup \mathcal{N}_S)$

- **Lemma 1**: Joint distribution of all disturbance terms $f(m_j, \forall j \in \mathcal{I}_1)$ is a multivariate Gaussian with covariance matrix C_m.
  - $C_{ii} = \sum_{j \neq i, j \in \mathcal{N} \setminus \mathcal{N}_C} \sigma_{ij}^2 + \sigma_i^2$
  - $C_{ij} = -\sigma_{ij}^2$ for $i \neq j$

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- **Lemma 2**: Privacy loss $\mathcal{L}_i$ for client i follows a Gaussian distribution.
  - Mean: $\frac{C_{ii}^{-1}}{2} ||\omega_i(\mathcal{D}_i) - \omega_i(\mathcal{D}_i')||_2^2$
  - Variance: $[\sum_{j \in \mathcal{I}_1} (C_{ij}^{-1})^2 C_{jj}] ||\omega_i(\mathcal{D}_i) - \omega_i(\mathcal{D}_i')||_2^2$

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- **Proposition 1 (Sufficient Conditions for (ε, δ)-DP)**:

$$\frac{1}{\sqrt{\sum_{j \in \mathcal{I}_1} (C_{ij}^{-1})^2 C_{jj}}} \geq \frac{\sqrt{2 \log(2/\delta)} \Delta}{\varepsilon}, \quad \forall i \in \mathcal{I}_1, \forall \mathcal{I}_1, \mathcal{I}_2$$

where Δ is the sensitivity of local parameters. This considers the worst case for unknown $\mathcal{N}_C, \mathcal{N}_S$.

Learning Convergence and Noise Optimization

- **Convergence Bound:** Conditional on the number of stragglers for all rounds $\mathbf{S} = \{S^{(t)}, t \in \mathcal{T}\}$, **Proposition 2** provides an upper bound for $\mathbb{E}\left\{F\left(\omega^{(T)}\right) - F\left(\omega^{*}\right)|\mathbf{S}\right\}$ which depends on the accumulated global noise $n_D^{(t)}$ at each round t.
- **Objective for noise optimization**: Minimize **expected** convergence bound:

$$\min \mathbb{E}_{S \sim g(s)} \mathbb{E}\left\{F\left(\omega^{(T)}\right) - F\left(\omega^{*}\right)|\mathbf{S}\right\}$$

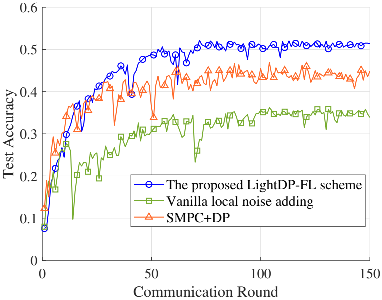
subject to DP constraints (already worst-case for any S distributions).

Experiment Setup

- **Dataset & Model**: CIFAR-10 with ResNet-18. Non-IID data distribution.
- **FL Setup**: FedAvg aggregation, $N = 50$ clients, $T = 150$ communication rounds.
- **Baselines for Comparison** (same DP level ε, δ):
  - Vanilla Local DP (LDP): Each client adds local noise.
  - SMPC + DP: e.g., Homomorphic Encryption (HE) + DP noise, SecAgg + DP noise, Coded Computation + DP noise.
- **Metric**: Test accuracy.

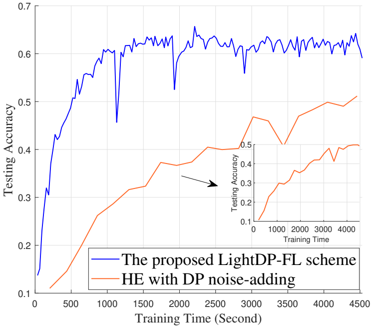
Accuracy vs. Communication Rounds

- Faster convergence and higher test accuracy ($> 15\%$ improvement) compared to vanilla LDP.
- Better accuracy than SMPC+DP schemes.
- **Reason**: SMPC+DP often relies on worst-case $\bar{C}$, $\bar{S}$ for noise addition, potentially adding excessive noise if actual $S < \bar{S}$. LightDP-FL optimizes noise based on the distribution of S, allowing for better performance.



Accuracy vs. Training Time

- Considers a small link failure probability (0.05).
- SecAgg and SwiftAgg+ (reliant on many C2C links) fail to train due to link fragility leading to too many stragglers.
- HE scheme suffers from significantly longer computation time per round, leading to slower convergence (by at least 10x) compared to LightDP-FL.



Privacy-Accuracy Trade-off

- **Impact of ε (Privacy Budget)**:
  - As $\varepsilon \downarrow$ (stricter privacy), noise $\uparrow$, accuracy $\downarrow$.
- **Impact of $\bar{C}$ (Max Colluders)**:
  - As $\bar{C} \uparrow$, noise $\uparrow$ (both individual and pairwise), accuracy $\downarrow$.
- **Impact of $\bar{S}$ (Max Stragglers)**:
  - As $\bar{S} \uparrow$, accuracy $\downarrow$. To maintain straggler resilience with higher likely S, pairwise noise variance σ_K^2 is kept lower, requiring higher σ_U^2 for DP, impacting accuracy.

Conclusion

- We proposed a private FL framework, **LightDP-FL**, which is:
  - Lightweight (low communication and computation loads).
  - Ensures provable Differential Privacy.
  - Resilient to straggling clients (in both client-to-server and client-to-client links).
  - Achieved by carefully designing the addition of both pairwise and individual Gaussian noise.

Thank You!